Project Progress Update

AgriCom, ESMT

March 27, 2026

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Executive Summary:

As we outlined in our last update, this week we completed the work for Phase 1 and 2 of our timeline. This work included the data collection and assembly as well as some early analysis as discussed with Karam in our last meeting. Following this we will begin with the testing of our slightly revised hypotheses that we shared in the last update, which change to account for known limitations based on data availability discussions with Karam. We look forward to your detailed thoughts on these pieces as well as on the previous report that we sent.

For clarity we also want to share an overview of the remaining meetings and deadlines we will have until the completion of this project:

- Monday March 30th, 4-5pm CET: ACP Bi-weekly meeting
- Friday April 3rd (morning), ESMT Team to send Final Update (which aims to mostly serve as a final alignment in case any outstanding questions remain before report submission)
- Sunday April 5th, Final Report completed
- Tuesday April 7th, Proposed Brief Check-In Meeting following report completion and ahead of final presentation
- Tuesday April 14th, 4:15-5:15pm CET, Final Presentation

Data Assembly:

The data collection phase is complete. We have assembled a panel of 273 weekly observations spanning January 2021 to March 2026, drawing on Google Trends search interest for the four target products, Berlin weather data from Open-Meteo, and macroeconomic indicators from Eurostat and FRED. All three of these sources are fully populated with less than 5% missingness, limited to the most recent weeks where Google Trends data lags. The one outstanding gap remains the pricing data, we have created a placeholder template that would be ready to receive the expected data.

To give a sense of the signal variation across products: Bio Salat shows the highest average search interest over the panel (mean: 69), while Bio Gurken is the most volatile (std dev: 30), reflecting its more intermittent seasonal demand. The table below highlights this variation:

Product	Mean GT Interest	Std Dev	Notes
Bio Tomaten	**65**	**17**	**Strong, consistent seasonality**
Bio Salat	**69**	**15**	**Highest average interest**
Bio Gurken	**40**	**30**	**High volatility, intermittent peaks**
Bio Paprika	**46**	**15**	**Stable, moderate interest**

Exploratory Analysis:

Search Trends & Macro Context

By plotting the five-year search time-series alongside key macro events, we observe Bio Tomaten's strong summer seasonality (peaks of 80–100, winter troughs of 20–30) holds consistently across all years, suggesting the signal is structurally reliable.

Key Correlations

Two findings from the correlation analysis stand out ahead of the formal hypothesis tests (see appendix). First, sunshine hours show a meaningful positive relationship with Bio Tomaten search interest ($r = 0.44$), which gives early support to H5 before we run the distributed lag models. Second, consumer confidence and the food price index move in opposite directions ($r = -0.60$), which effectively captures the inflationary squeeze of 2022–23 in our data. This macro dynamic will be important context when interpreting the price elasticity results in H3.

Cross-product correlations are also notable: Bio Tomaten and Bio Gurken are closely linked ($r = 0.70$), pointing to shared weather-driven demand patterns as we discussed previously, whereas Bio Paprika moves more independently, something we will explore further in the hypothesis testing phase.

Seasonal Structure

Seasonal decomposition confirms that Bio Tomaten's annual cycle accounts for the bulk of its variation, with the residual component remaining small throughout the panel.

Regarding the pricing data

To reiterate, we would be grateful to receive the weekly Berlin pricing data at your earliest convenience.

Our Request:

- Four specific SKUs, organic and conventional variants of each: Round tomatoes (loose or standard pack), bagged leafy greens / mixed salad, cucumbers (standard full-size, not mini cucumbers), and bell peppers
- Price per kilogram in EUR (€/kg), consistently applied across all products and both variants
- Berlin region
- Minimum 52 weeks of history, ideally 104 weeks (two full years). Two years is preferred because it covers two complete seasonal cycles, which is essential for the seasonal interaction terms in H3 and for isolating the price elasticity from seasonal demand fluctuations. A single year risks conflating a price movement with a seasonal movement if the two happen to coincide.

This would materially strengthen the price elasticity estimation in H3 and improve the precision of the price component in the demand index. If the data is unable to be provided before the end of day, **March 27th**, we will proceed using proxy data as the fallback, so as to ensure that the project will not be delayed. The final report will indicate clearly which source was used and how this affects the results.

Appendix

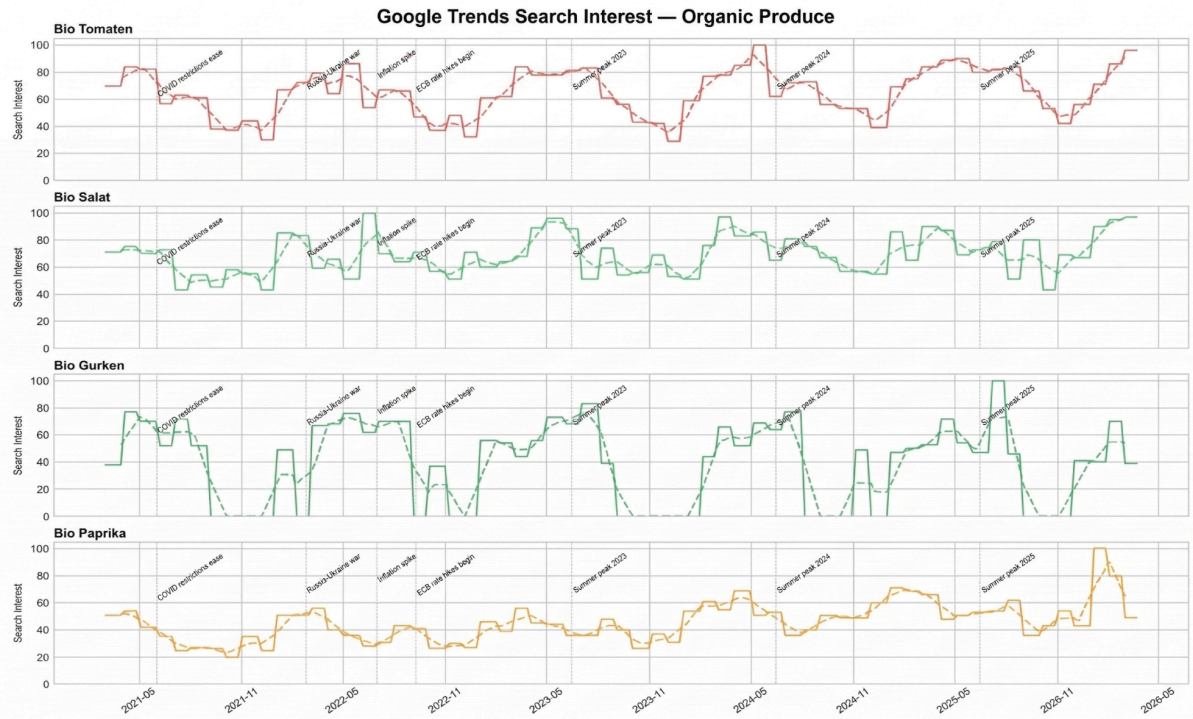

Figure 1: Google Trends search interest for organic produce keywords (2021-2026), annotated with macroeconomic events.

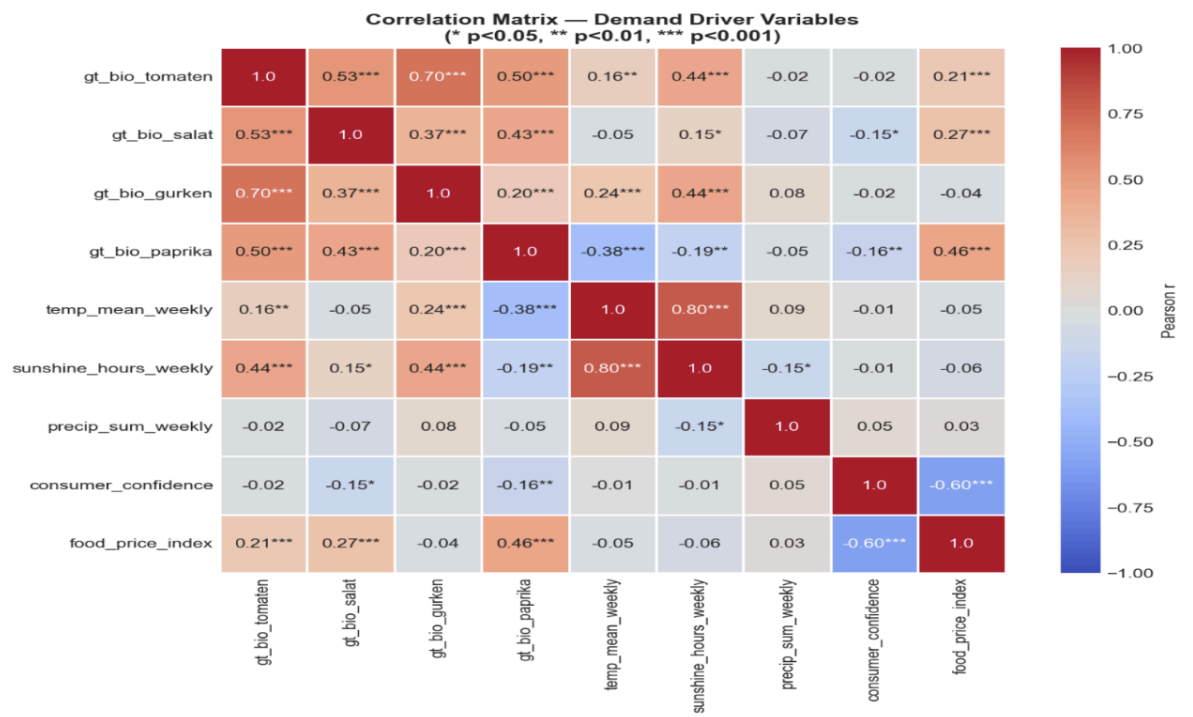

Figure 2: Pearson correlation heatmap across demand driver variables (p<0.05, ** p<0.01, *** p<0.001).*